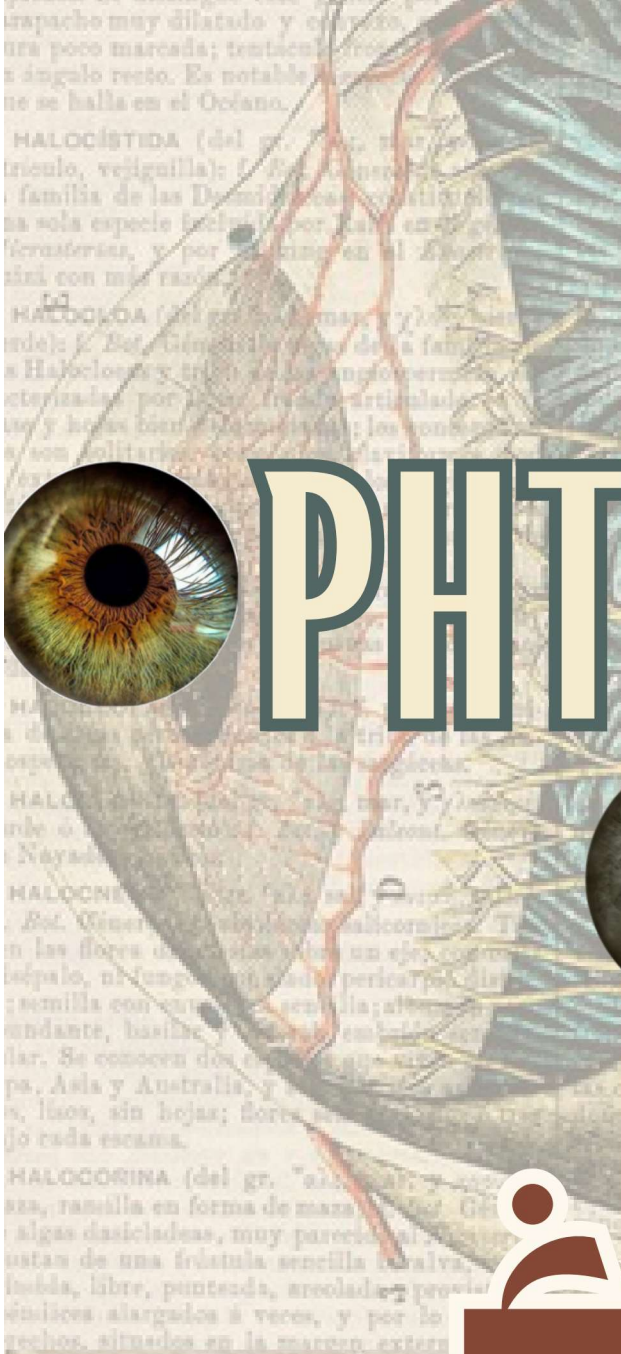




OPHTHALMOLOGY

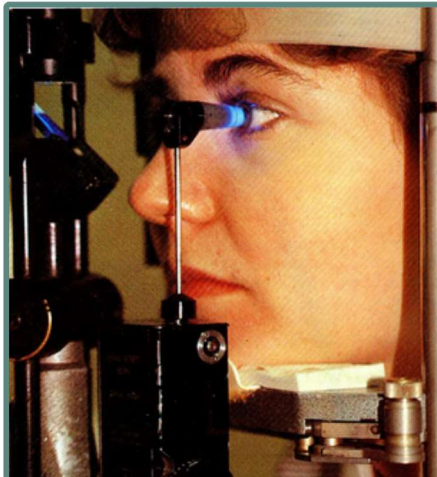
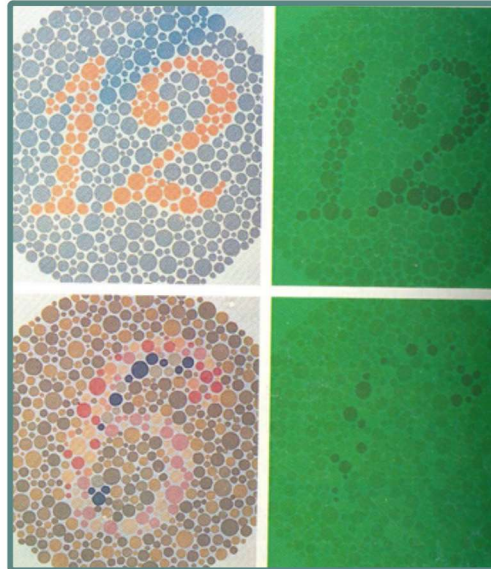
مختار
ميتنج العملي

SUPPORT 43



METHODS OF EXAMINATION

- بسم الله توكلت على الله ولا حول ولا قوة الا بالله
- العملي 30 سلايد باذن الله بعضهم فقط spot diagnosis والبعض معاها أسئلة نظري هنجمعهم في الفايل هنا باذن الله



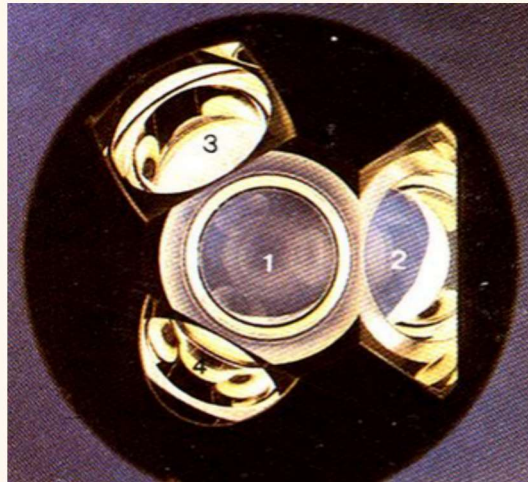
قال صلى الله عليه وسلم : (من كانت الآخرة همّة جعل الله غناه في قلبه وجمع له شمله وأتته الدنيا وهي راغمة، ومن كانت الدنيا همّة جعل الله فقره بين عينيه وفرّق عليه شمله، ولم يأت من الدنيا إلا ما قُدّر له)



fundus examination by
direct ophthalmoscope



fundus examination by
indirect ophthalmoscope



العصفور
بيقولك اكتب
الوظيفتين تخذ
درجتك



Goldman 3 mirror contact lens used for
fundus examination and gonioscopy

Schoitz
indentation
Tonometry for
IOP
measurement



Schoitz disadvantage

- False low readings when **scleral rigidity is low** as in: high myopia and previous ocular surgery.



EYE LID

Preseptal
cellulitis

- Interpret the attached CT findings?

- opacification at anterior orbital septum
- Thickening of eye lid soft tissues
- Inflammation confined anterior to the orbital septum (the orbit is normal)
- Intact ocular motility
- No proptosis
- No orbital fat stranding+No subperiosteal abscess

ازاي افرقه عن
ال orbital
cellulitis

- Sources of infection?

- **Endogenous:** extension from neighboring structures as the paranasal sinuses.
- **Exogenous:** as in penetrating trauma or septic orbital surgery.
- **Blood born:** metastatic infection as in pyemia.

- Treatment? (same as orbital cellulitis)

- **Hospitalization:** is necessary in orbital cellulitis only for close follow-up:
 - 1) Topical and systemic parenteral broad spectrum antibiotics.
 - 2) Hot foment.
- **N.B:** If abscess is formed, it should be drained



Acute orbital cellulitis



Herpes simplex

- **C.O:** Herpes simplex type 1



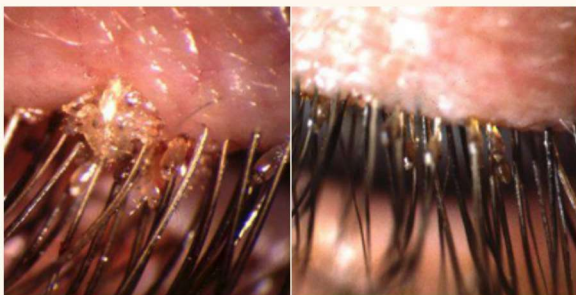
Herpes zoster ophthalmicus

- **C.O:** herpes zoster virus
- What is Hutchinson sign?
 - Vesicular lesions on the tip of the nose in HZO, indicating nasociliary nerve involvement (affected parts are along its distribution)
- Name specific drug used for treatment (**acyclovir "Zovirax"**)



Ulcerative blepharitis

- **C.O:** Staph. aureus



Parasitic blepharitis

- **C.O:** Infestation by Pubic crab louse and its ova

Scaly blepharitis



Treatment

- **General TTT:** TTT of predisposing factors & associated conditions.
- **Local TTT:** Lid hygiene using:
 - Warm compresses, Eyelid massage.
 - Removal of scales using diluted solution of baby shampoo.
 - Local antibiotic ointment.



Stye: Zeiss glands



كلما كان العبد حسن الظن بالله حسن الرجاء له صادق التوكل عليه: فإن الله لا يخيب أمله فيه البتة، فإنه سبحانه لا يخيب أمل آمل ولا يضيع عمل عامل.

ابن القيم رحمه الله - الفوائد

Infected chalazion



Treatment

- TTT of the predisposing factors.
- Warm, moist compresses and topical antibiotics (drops and ointment).
- Systemic antibiotics are indicated in severe cases, complicated by cellulites.
- If Pointing: drain pus by epilation of the related eye lash with a horizontal incision.
- Do not press to evacuate a stye (dangerous area) may spread infection and cause orbital cellulites or cavernous sinus thrombosis.
- **Infected chalazion has Same TTT for stye**, but evacuation may be needed through a vertical or cruciate incision on the conjunctival side.



Chalazion: meibomian gland

Treatment

- **Small chalazion:** may be left alone with follow up.
- **Large chalazion:** incision (cruciate or vertical) and curettage or scraping of its contents from the conjunctival side
- **Infected chalazion:** As hordeolum internum.
- **Recurrent swelling in the same gland:** Excision biopsy and histopathological examination is done to exclude malignancy



EYELASHES ABNORMALITIES

Trichiasis



Trichiasis

- **Def:** This is an acquired condition in which more than 4 lashes are directed posteriorly and rub against the cornea or bulbar conjunctiva

Distichiasis



Distichiasis

- **Def:** an abnormality (usually congenital) in which extra row of lashes arise from the lid margin behind the grey line in the place of Meibomian gland orifices.



Trichiasis

Complication

- Chronic conjunctivitis, conjunctival ulceration & epithelial plaque.
- Corneal: recurrent ulceration which may be followed by opacification, superficial vascularization & epithelial plaque.

Madarosis

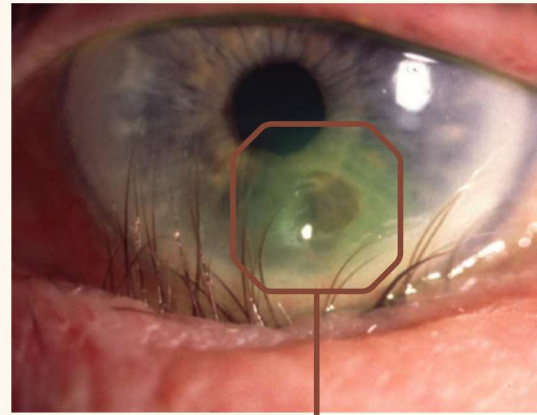
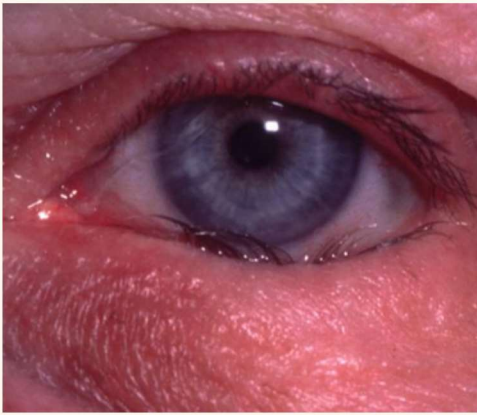


Madarosis

Poliosis



Poliosis



Involutional entropion

- Corneal ulcer stained with Fluorescein



Cicatricial entropion



Causes

- 1) Trachoma (the commonest cause in Egypt)
- 2) Ulcerative blepharitis.
- 3) Trauma and Chemical burns.
- 4) Cicatrizing diseases: ocular cicatricial pemphigoid and Stevens Johnson disease.



تَتَوَقَّى قَبْلَ الرِّحِيلِ الرِّحِيلَا
أَنْ تَرَى فَوْقَهَا النَّدَى إِكْلِيلَا

إِنَّ سَرَّ الْجَنَّةِ فِي الْأَرْضِ نَفْسُ
وَتَرَى الشَّوْكَ فِي الْوُرُودِ وَتَعْمَى



Involucional ectropion



Mechanism

- due to age related degeneration of elastic tissue within the eyelid leading to horizontal **laxity** .
- Overriding of pre-tarsal over pre-septal orbicularis oculi and vertical eyelid instability due to weak or dis-inserted lower eyelid retractors are contributing factors.

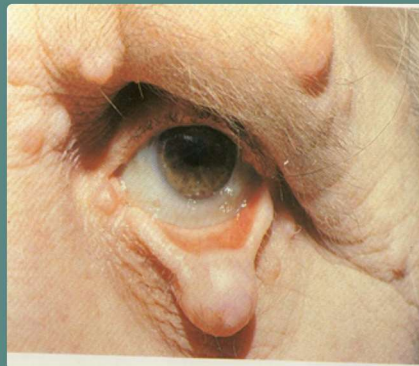
Cicatracial ectropion



Treatment

- Protect the cornea from exposure until the deformity is corrected surgically by:
 - **Small scar:** V-Y operation.
 - **Large scar:** removal of the scar tissue followed by eyelid grafting.

Mechanical ectropion

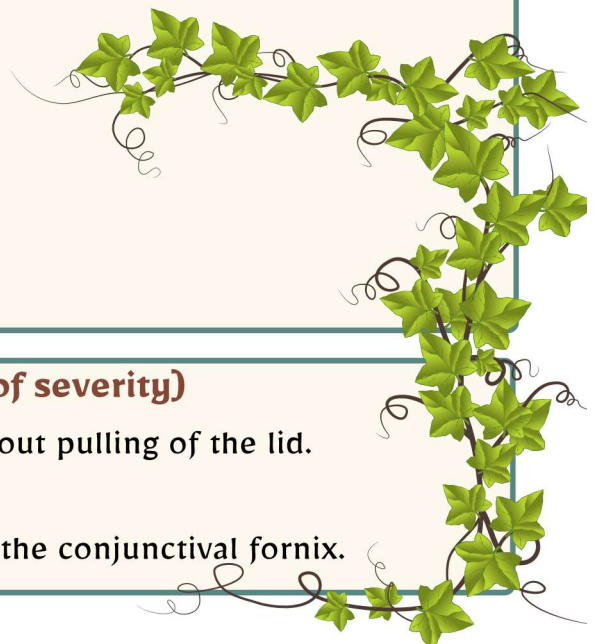


Treatment

- Remove of cause

Grades of Ectropion (Grading of severity)

- **Mild:** In which the lacrimal punctum is visible without pulling of the lid.
- **Moderate:** The tarsal conjunctiva is exposed.
- **Severe:** A completely everted lid with exposure of the conjunctival fornix.





Paralytic ectropion & lagophthalmos & epiphora

• **Tested Nerve:** Fascial nerve

• **Tested muscle:** Orbicularis oculi



Paralytic ectropion & lagophthalmos

• **Tested Nerve:** Fascial nerve

• **Tested muscle:** Orbicularis oculi



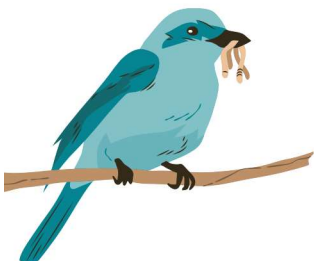
Paralytic ectropion & lagophthalmos

• **Tested Nerve:** Fascial nerve

• **Tested muscle:** Orbicularis oculi

Complication of lagophthalmos

- **Cornea:** Exposure keratitis develops which may end in corneal ulceration, perforation and loss of the eye.
- **Conjunctiva:** may be exposed, dry and xerotic.



﴿صُنِعَ اللَّهُ الَّذِي أَتَقَنَ كُلَّ شَيْءٍ﴾





mention two sign (facial nerve injury)

- | | | |
|--|---|----------------------------|
| • Loss of forehead wrinkles | + | Inability to close eyes |
| • Wide palpebral fissure | + | Epiphora |
| • Loss of naso-labial fold | + | Drooping of angle of mouth |
| • Dribbling of food while chewing on affected side | | |



Xanthelasma



Sturge Weber syndrome
(Nevus flammeus)

Ocular associations

- Congenital Glaucoma
- Choroidal hemangioma



Capillary hemangioma



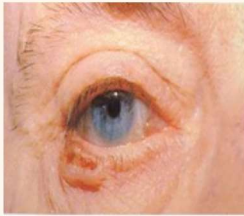
Periocular hemangioma



Nevus



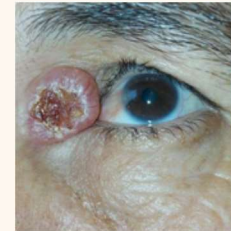
Melanoma



Basal cell carcinoma
(Rodent ulcer)



SCC



Basal cell carcinoma
+ Pterygium



Lt. Ptosis



Lid retraction



Lt. Ptosis

Management

Depends on Amount of ptosis and levator action:

- 1) If levator muscle has some action
(fair or good): في الصورة هنا غالبا كده
 - Resection and advancement done to strengthen the muscle.
- 2) If Levator function is **poor & severe ptosis**:
 - Frontalis suspension (sling) (Hiss operation).



Unilateral severe ptosis

Management

زي الي جنبه بس الحالة اصعب فهنلجاً لرقم 2
Frontalis suspension

When to manage? and why?

- Congenital ptosis: Age of interference:

- Mild to moderate: Wait till the school age (5-6 years).
- Severe: as early as possible **to avoid amblyopia**. النقطة دي هي الاجابة هنا



pseudoptosis due to



Lack of lid support

تكتب ان فيه
pseudoptosis
وضرووري تكتب السبب

Contralateral lid
retractionIpsilateral
hypotropiaBrow ptosis
excessive eyebrow skinDermatochalasis
excessive eyelid skin

الصورة دي مهمة

Marcus Gunn jaw - winking syndrome



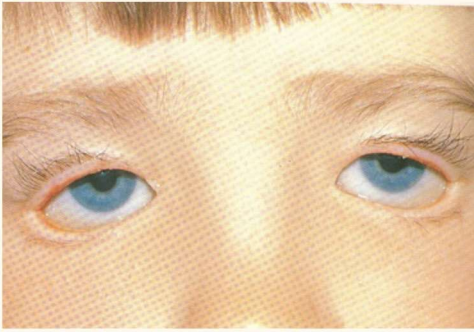
Opening of mouth



Contralateral movement of jaw

- **Cause:** misinnervation between CN V (mandibular branch of trigeminal) and CN III (oculomotor), leading to eyelid elevation with jaw movement.





Bilateral congenital upper & lower lid ectropion (blepharophimosis syndrome)

Components of this syndrome

- Moderate to severe symmetrical ptosis
- Short horizontal palpebral aperture
- Telecanthus (lateral displacement of medial canthus)
- Epicanthus inversus (lower lid fold larger than upper)
- Lateral inferior ectropion
- Poorly developed nasal bridge and hypoplasia of superior orbital rims



Left third nerve palsy



- **Procedure:** Frontalis suspension
- **Indication:** If Levator function is poor & severe ptosis



Horner syndrome (Rt. eye)

Signs

- mild unilateral ptosis, miosis and anhidrosis, enophthalmos



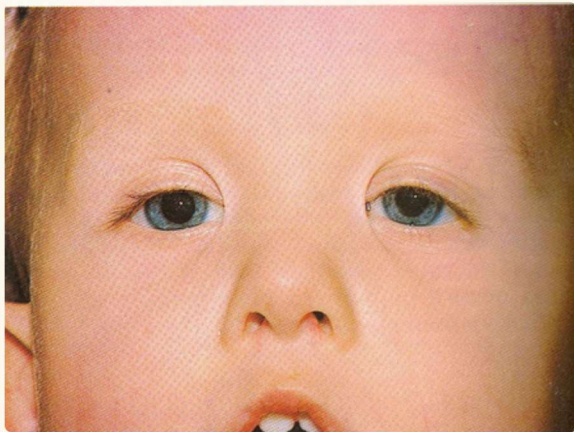
Bilateral involutional Ptosis

Pathogenesis

- Due to Dehiscence (degenerative changes) in levator muscle aponeurosis while the levator function is good.

Why there is high upper lid crease?

- The aponeurosis disinserts or stretches from the tarsal plate but its remaining attachments to the skin remain. more impulses sent to the muscle to correct ptosis lead to elevation of the crease



Epicanthus



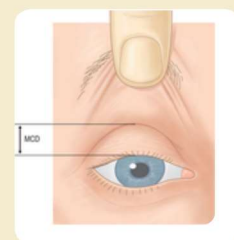
Rt. Upper lid coloboma

Complications

- Exposure keratopathy
- Corneal ulcer

MARGIN CREASE DISTANCE

- **Normal:** 7-9 mm in males, 8-10 mm in females
- **Absent** lid crease in severe congenital ptosis
- **High** lid crease in aponeurotic ptosis

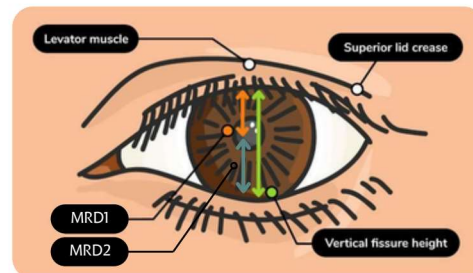
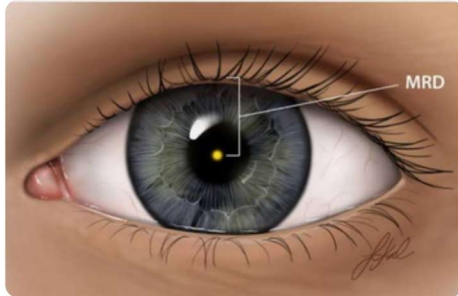


الحته دي كانت المفروض الصفحة الجاية بس مفيهاش مكان



EYE LID TESTS

MARGIN TO REFLEX DISTANCE (MRD) TEST



ممکن یشاور زی الصور علی النقطة بین ال pupil لل center of lid margin
تم یسأل هی عبارة عن ایه وایه القيمة الطبيعية

- **Distance:** Margin to reflex distance 1
- **Normal value:** 4-5 mm
- **Normal MRD2 :** >5 mm

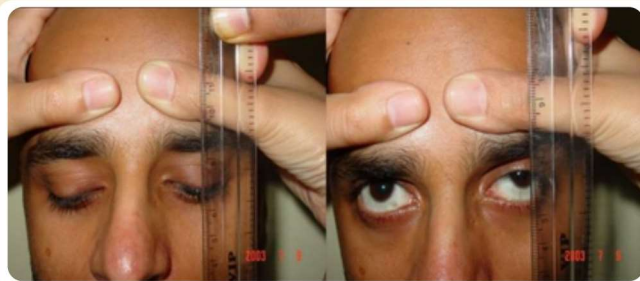
انت بتشوف قل اد ایه Interpretation مش بقى کم فاهمني؟

Drop by 1 mm
Mild ptosis

Drop by 2-3 mm
Moderate ptosis

Drop by >4 mm
Severe ptosis

LEVATOR FUNCTION TEST



- **Use:** Determine the levator function power
- **Normal vertical Palpebral fissure :** 9-10 mm

Interpretation

≥ 12 mm
Excellent

8-11 mm
Good

5 – 7 mm
Fair

0 -4 mm
Poor